Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 5 Marks

NO. OF FUNCTIONAL FEATURES TEMPLATE

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Web Input Form	Renders form to capture 12 applicant parameters.	Frontend (home.html)	Done	1 Mark
2	Policy Engine	Evaluates income floors and student risk override rules.	Backend (app.py)	Done	1 Mark
3	Feature Transformer	Calculates ratios and scales variables using scaler.pkl.	Backend (model.py)	Done	1 Mark
4	Random Forest scoring	Scores scaled inputs using calibrated threshold.	Backend (credit_model.pkl)	Done	1 Mark
5	Outcomes Dashboard	Stylized approval/rejection decision panels and logs.	Frontend (result.html)	Done	0.5 Marks
6	Static EDA Charts	Displays distribution plots of training datasets.	Frontend UI	Done	0.5 Marks

Feature Summary Metrics

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	1
Features Tested & Verified	6

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Input form page, Outcomes decision banners, progress risk bars, EDA charts.	home.html, result.html, CSS theme
2	Backend / Logic	Flask request handlers, Layer 1 policy validator, feature converter.	app.py, engineer_features()
3	Database / Storage	Fitted scaling weights, calibrated thresholds, Random Forest coefficients.	scaler.pkl, threshold.pkl, credit_model.pkl
4	API / Integration	Post endpoint processing form data and returning JSON templates.	Flask predict route (/predict)
5	Security / Audit	Input boundaries validation, policy audit logging, PII omission.	Try-except blocks, policy log tables